Supplementary Information

A Novel Double-Conical Heat Source for Fast and Accurate Prediction of Multi-Track Melt Pool Geometry and Cooling Rates in Laser Powder Bed Fusion: NIST AM-Bench 2025 Challenge

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Overview

This Supplementary Information document describes the electronic supplementary materials accompanying the manuscript. The supplementary package contains 23 files: seven transient simulation videos, seven temperature-history videos, eight spreadsheet files containing supporting numerical data, and one image illustrating the steady-state condition. In the manuscript and the journal submission system, these materials are cited consecutively as *Online Resource 1* through *Online Resource 23*.

The supplementary materials provide additional visualization and numerical data supporting the model development, calibration, validation, and analysis presented in the manuscript. Detailed descriptions of the individual online resources are provided in the following sections.

Summary of Supplementary Files

Resource	File Type	Case	Content
ESM_1	Video (MP4)	AM-Bench 2022, single track	Transient simulation using the single-conical heat-source model.
ESM_2	Video (MP4)	AM-Bench 2022, single track	Transient simulation using the Double Conical Heat Source (DCHS) model.

Resource	File Type	Case	Content
ESM_3	Video (MP4)	AM-Bench 2022, XPad	Transient 10-track simulation using the single-conical heat-source model.
ESM_4	Video (MP4)	AM-Bench 2022, XPad	Transient 10-track simulation using the DCHS model.
ESM_5	Video (MP4)	AM-Bench 2025, single track	Transient DCHS simulation used for single-track model calibration.
ESM_6	Video (MP4)	AM-Bench 2025, Challenge 7	Transient 15-track DCHS simulation with 5 ms turnaround time.
ESM_7	Video (MP4)	AM-Bench 2025, Challenge 7	Transient 15-track DCHS simulation with 0.75 ms turnaround time.
ESM_8	Video (MP4)	AM-Bench 2022, single track	Temperature history for the single-conical heat-source simulation.
ESM_9	Video (MP4)	AM-Bench 2022, single track	Temperature history for the DCHS simulation.
ESM_10	Video (MP4)	AM-Bench 2022, XPad	Temperature histories for the single-conical multi-track simulation.
ESM_11	Video (MP4)	AM-Bench 2022, XPad	Temperature histories for the DCHS multi-track simulation.
ESM_12	Video (MP4)	AM-Bench 2025, single track	Temperature history for the DCHS single-track simulation.
ESM_13	Video (MP4)	AM-Bench 2025, Challenge 7	Temperature histories for the 5 ms turnaround-time case.
ESM_14	Video (MP4)	AM-Bench 2025, Challenge 7	Temperature histories for the 0.75 ms turnaround-time case.
ESM_15	Spreadsheet (XLSX)	AM-Bench 2022, XPad	Pad depth, overlap depth, width, and overlap-width data and comparison.
ESM_16	Spreadsheet (XLSX)	AM-Bench 2025, Challenge 7	Experimental and simulated pad dimensions for the 5 ms and 0.75 ms cases.
ESM_17	Spreadsheet (XLSX)	AM-Bench 2022, single track	SCR, TCR, LCR, and TAM for the single-conical and DCHS models.
ESM_18	Spreadsheet (XLSX)	AM-Bench 2022, XPad	Point-wise TAM and SCR data for the single-conical and DCHS models.
ESM_19	Spreadsheet (XLSX)	AM-Bench 2025, single track	Point-wise SCR, TCR, LCR, and TAM from the DCHS simulation.
ESM_20	Spreadsheet (XLSX)	AM-Bench 2025, Challenge 7	Point-wise TAM and SCR data for the 5 ms and 0.75 ms turnaround cases.
ESM_21	Spreadsheet (XLSX)	AM-Bench 2022, single track	Grid-independence results based on geometric and thermal metrics.
ESM_22	Spreadsheet (XLSX)	AM-Bench 2022/2025	Coordinates of monitoring points used for temperature-history extraction.

Resource	File Type	Case	Content
ESM_23	Spreadsheet (XLSX)	AM-Bench 2022/2025	Summary of process conditions, heat-source parameters, and heat-transfer parameters used for all simulation cases.
ESM_24	Image (PNG)	Single-track simulation	Melt-pool dimensions versus laser travel distance demonstrating steady state.
ESM_25	PDF	All	Mesh images.

S1. Transient Simulation Videos

Online Resources 1–7 provide transient contour animations of the numerical simulations presented in the manuscript. Each video includes contours of temperature, liquid fraction, volumetric laser heat-source intensity, radiative heat-loss source term, and convective heat-loss source term. Temperature is reported in K, liquid fraction is dimensionless, and the volumetric heat-source intensity and heat-loss source terms are reported in W/m^3.

For each field, contours are presented in the x–z plane, representing a longitudinal section along the laser scanning direction, and in the x–y plane, representing the top view of the build surface. The videos illustrate the temporal evolution of the thermal field, melt-pool formation and solidification, applied volumetric heat source, and heat losses during laser scanning. The opening screen of each video provides the corresponding simulation conditions and numerical parameters, including mesh information, time-step size, track length, laser power, scan speed, and other relevant case-specific parameters.

Online Resource 1 (ESM_1.mp4).
Transient thermal simulation of the NIST AM-Bench 2022 single-track Case 0 using the conventional single-conical volumetric heat-source model. This simulation is used in the single-track validation presented in the main manuscript and serves as the reference heat-source model for comparison with the proposed Double Conical Heat Source (DCHS). The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, and a track length of 2 mm. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 2 (ESM_2.mp4).
Transient thermal simulation of the NIST AM-Bench 2022 single-track Case 0 using the proposed Double Conical Heat Source (DCHS) model. This simulation is used for the single-track validation of the proposed model in the main manuscript and is directly compared with the single-conical heat-source simulation provided in Online Resource 1. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms. The simulation uses the same process conditions as Online Resource 1, with a laser power of 285 W, a scan speed of 960 mm/s, and a track length of 2 mm. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 3 (ESM_3.mp4).

Transient thermal simulation of the NIST AM-Bench 2022 XPad multi-track benchmark using the conventional single-conical volumetric heat-source model. This simulation is used in the multi-track XPad validation presented in the main manuscript and serves as the reference case for comparison with the proposed Double Conical Heat Source (DCHS) model. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during the simulation of 10 bidirectional tracks. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 2 mm, and a hatch spacing of 110 μm. Laser turnaround times of 5 ms and 40 ms were applied after odd- and even-numbered tracks, respectively. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 4 (ESM_4.mp4).
Transient thermal simulation of the NIST AM-Bench 2022 XPad multi-track benchmark using the proposed Double Conical Heat Source (DCHS) model. This simulation is used in the multi-track XPad validation of the proposed model presented in the main manuscript and is directly compared with the conventional single-conical heat-source simulation provided in Online Resource 3. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during the simulation of 10 bidirectional tracks. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 2 mm, and a hatch spacing of 110 μm. Laser turnaround times of 5 ms and 40 ms were applied after odd- and even-numbered tracks, respectively. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 5 (ESM_5.mp4).
Transient thermal simulation of the NIST AM-Bench 2025 single-track case using the proposed Double Conical Heat Source (DCHS) model. This single-track case was provided as part of the AM-Bench 2025 competition and was used to calibrate the DCHS model prior to its application to the multi-track Challenge 7 cases presented in the main manuscript. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during laser scanning. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 2 mm, and a laser spot diameter of 72 μm. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 6 (ESM_6.mp4).
Transient thermal simulation of the NIST AM-Bench 2025 Challenge 7 multi-track case with a laser turnaround time of 5 ms using the proposed Double Conical Heat Source (DCHS) model. This case constitutes one of the primary multi-track prediction cases of the AM-Bench 2025 competition and was simulated using the DCHS model calibrated from the corresponding single-track case provided in Online Resource 5. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during 15 bidirectional tracks. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 1 mm, a hatch spacing of 110 μm, and a laser spot diameter of 72 μm. A turnaround time of 5 ms was applied between consecutive tracks. Detailed computational and processing parameters are provided on the opening screen of the video.

Online Resource 7 (ESM_7.mp4).

Transient thermal simulation of the NIST AM-Bench 2025 Challenge 7 multi-track case with a laser turnaround time of 0.75 ms using the proposed Double Conical Heat Source (DCHS) model. This case constitutes the second primary multi-track prediction case of the AM-Bench 2025 competition and was simulated using the DCHS model calibrated from the corresponding single-track case provided in Online Resource 5. The video presents the transient evolution of the temperature field, liquid fraction, volumetric laser heat-source intensity, and radiative and convective heat-loss source terms during 15 bidirectional tracks. The simulation was performed with a laser power of 285 W, a scan speed of 960 mm/s, a track length of 1 mm, a hatch spacing of 110 μm, and a laser spot diameter of 72 μm. A turnaround time of 0.75 ms was applied between consecutive tracks. Detailed computational and processing parameters are provided on the opening screen of the video.

S2. Temperature-History Videos

Online Resources 8–14 present the transient temperature histories extracted from the corresponding single-track and multi-track simulations. These videos illustrate the heating and cooling cycles experienced at the selected monitoring locations as the laser passes over or near each point. The temperature histories are used to evaluate the thermal metrics reported in the main manuscript, including the Solid Cooling Rate (SCR), Transition Cooling Rate (TCR), Liquid Cooling Rate (LCR), and Time Above Melt (TAM) for the single-track cases, as well as the Pad Solid Cooling Rate (PSCR) and Pad Time Above Melt (PTAM) for the multi-track cases.

The horizontal threshold lines shown in the temperature-history plots indicate the temperature ranges used to evaluate the corresponding thermal metrics. Temperature is reported in K and time in seconds. The animations show the evolution of the temperature history as the transient simulation progresses and provide a visual representation of the thermal cycles used for the quantitative analyses presented in the main manuscript.

Online Resource 8 (ESM_8.mp4).
Temperature-history animation for the NIST AM-Bench 2022 single-track Case 0 simulated using the conventional single-conical heat-source model. The temperature history corresponds to the single-track validation analysis presented in the main manuscript and is used to evaluate the associated cooling-rate and time-above-melt metrics.

Online Resource 9 (ESM_9.mp4).
Temperature-history animation for the NIST AM-Bench 2022 single-track Case 0 simulated using the proposed Double Conical Heat Source (DCHS) model. The temperature history corresponds to the DCHS single-track validation presented in the main manuscript and is used to evaluate the associated Solid Cooling Rate (SCR), Transition Cooling Rate (TCR), Liquid Cooling Rate (LCR), and Time Above Melt (TAM). This resource provides the thermal-history counterpart to the transient simulation presented in Online Resource 2.

Online Resource 10 (ESM_10.mp4).
Temperature-history animation for the NIST AM-Bench 2022 XPad multi-track benchmark simulated using the conventional single-conical heat-source model. The temperature histories correspond to the XPad multi-track validation presented in the main manuscript and illustrate the repeated thermal cycles produced by successive laser tracks. The histories are used to determine the Pad Time Above Melt (PTAM) and Pad Solid

Cooling Rate (PSCR). This resource provides the temperature-history counterpart to the transient simulation presented in Online Resource 3.

Online Resource 11 (ESM_11.mp4).
Temperature-history animation for the NIST AM-Bench 2022 XPad multi-track benchmark simulated using the proposed Double Conical Heat Source (DCHS) model. The temperature histories correspond to the DCHS XPad multi-track validation presented in the main manuscript and illustrate the repeated heating and cooling cycles caused by successive laser tracks and remelting. These histories are used to determine the Pad Time Above Melt (PTAM) and Pad Solid Cooling Rate (PSCR). This resource provides the temperature-history counterpart to the transient DCHS simulation presented in Online Resource 4.

Online Resource 12 (ESM_12.mp4).
Temperature-history animation for the NIST AM-Bench 2025 single-track case simulated using the proposed Double Conical Heat Source (DCHS) model. This single-track case was used to calibrate the DCHS model prior to its application to the AM-Bench 2025 Challenge 7 multi-track prediction cases presented in the main manuscript. The animation illustrates the transient heating and cooling response at the selected monitoring location and the temperature thresholds used to evaluate the corresponding thermal metrics. This resource provides the temperature-history counterpart to the transient simulation presented in Online Resource 5.

Online Resource 13 (ESM_13.mp4).
Temperature-history animation for the NIST AM-Bench 2025 Challenge 7 multi-track case with a laser turnaround time of 5 ms, simulated using the proposed Double Conical Heat Source (DCHS) model. This case is one of the primary multi-track prediction cases of the AM-Bench 2025 competition. The animation illustrates the repeated heating, remelting, and cooling cycles at the selected monitoring locations during successive laser tracks. The temperature histories are used to evaluate the Pad Time Above Melt (PTAM) and Pad Solid Cooling Rate (PSCR) reported in the main manuscript. This resource provides the temperature-history counterpart to the transient simulation presented in Online Resource 6.

Online Resource 14 (ESM_14.mp4).
Temperature-history animation for the NIST AM-Bench 2025 Challenge 7 multi-track case with a laser turnaround time of 0.75 ms, simulated using the proposed Double Conical Heat Source (DCHS) model. This case is the second primary multi-track prediction case of the AM-Bench 2025 competition. The animation illustrates the repeated heating, remelting, and cooling cycles at the selected monitoring locations during successive laser tracks. The temperature histories are used to evaluate the Pad Time Above Melt (PTAM) and Pad Solid Cooling Rate (PSCR) reported in the main manuscript. This resource provides the temperature-history counterpart to the transient simulation presented in Online Resource 7.

S3. Supporting Numerical Data

Online Resource 15 (ESM_15.xlsx).
Spreadsheet containing the simulated melt-pool and pad-dimension data for the NIST AM-Bench 2022 XPad multi-track case. The file provides track-wise depth, overlap depth, width, and overlap width predicted using both the conventional single-conical and double conical heat source models. It also includes comparisons of the predicted pad geometry and thermal metrics with the experimental measurements, together with the

corresponding percentage errors and experimental standard deviations. The dataset supports the AM-Bench 2022 XPad multi-track validation and comparison presented in the main manuscript.

Online Resource 16 (ESM_16.xlsx).
Spreadsheet containing the experimental and simulated pad-dimension data for the NIST AM-Bench 2025 Challenge 7 multi-track cases with laser turnaround times of 0.75 ms and 5 ms. The file provides the pad depth, overlap depth, width, and overlap width at the measurement locations for comparison between the experimental measurements and predictions obtained using the Double Conical Heat Source (DCHS) model. These data support the validation of the AM-Bench 2025 multi-track predictions presented in the main manuscript.

Online Resource 17 (ESM_17.xlsx).
Spreadsheet containing the point-wise thermal metrics obtained from the NIST AM-Bench 2022 single-track Case 0 simulations using the conventional single-conical and proposed Double Conical Heat Source (DCHS) models. Separate worksheets provide the Solid Cooling Rate (SCR), Transition Cooling Rate (TCR), Liquid Cooling Rate (LCR), and Time Above Melt (TAM) calculated at the selected monitoring points for each heat-source model. These data support the single-track thermal validation and comparison of the two heat-source models presented in the main manuscript.

Online Resource 18 (ESM_18.xlsx).
Spreadsheet containing the point-wise thermal metrics obtained from the NIST AM-Bench 2022 XPad multi-track simulations using the conventional single-conical and proposed Double Conical Heat Source (DCHS) models. The file provides the Time Above Melt (TAM) and Solid Cooling Rate (SCR) calculated at the selected monitoring points for both heat-source models. These data were used to evaluate the Pad Time Above Melt (PTAM) and Pad Solid Cooling Rate (PSCR) and support the multi-track thermal validation and comparison presented in the main manuscript.

Online Resource 19 (ESM_19.xlsx).
Spreadsheet containing the point-wise thermal metrics obtained from the NIST AM-Bench 2025 single-track simulation using the proposed Double Conical Heat Source (DCHS) model. The file provides the Solid Cooling Rate (SCR), Transition Cooling Rate (TCR), Liquid Cooling Rate (LCR), and Time Above Melt (TAM) calculated at the selected monitoring points. These data characterize the thermal response of the calibrated single-track model used before its application to the AM-Bench 2025 Challenge 7 multi-track prediction cases.

Online Resource 20 (ESM_20.xlsx).
Spreadsheet containing the point-wise thermal metrics obtained from the NIST AM-Bench 2025 Challenge 7 multi-track simulations with laser turnaround times of 5 ms and 0.75 ms using the proposed Double Conical Heat Source (DCHS) model. Separate worksheets provide the monitoring-point coordinates, Time Above Melt (TAM), and Solid Cooling Rate (SCR) for the two turnaround-time cases. These data were used to determine the Primary Time Above Melt (PTAM) and Primary Solid Cooling Rate (PSCR) and to evaluate the influence of laser turnaround time on the multi-track thermal response presented in the main manuscript.

Online Resource 21 (ESM_21.xlsx).
Spreadsheet containing the grid-independence study for the NIST AM-Bench 2022 single-track Case 0

simulation using the proposed Double Conical Heat Source (DCHS) model. The file compares four computational meshes with different grid resolutions and cell counts. Melt-pool depth, width, area, maximum temperature, Solid Cooling Rate (SCR), Transition Cooling Rate (TCR), Liquid Cooling Rate (LCR), and Time Above Melt (TAM) are reported for each mesh. These results were used to evaluate mesh sensitivity and select an appropriate grid resolution for the numerical simulations presented in the main manuscript.

Online Resource 22 (ESM_22.xlsx).
Spreadsheet containing the coordinates of the monitoring points used to extract temperature histories from the numerical simulations. Separate worksheets provide the monitoring-point coordinates for the NIST AM-Bench 2025 multi-track, AM-Bench 2022 multi-track, and AM-Bench 2022/2025 single-track cases. The extracted temperature histories were used to plot the cooling curves and calculate the cooling-rate and time-above-melt thermal metrics reported in the main manuscript. The x, y, and z coordinates are reported in meters (m).

Online Resource 23 (ESM_23.xlsx).
Spreadsheet summarizing the principal process parameters and heat-source model parameters used in the numerical simulations presented in the manuscript. The file includes the NIST AM-Bench 2022 and AM-Bench 2025 single-track and multi-track cases, including the 5 ms and 0.75 ms turnaround-time cases. For each simulation, the spreadsheet identifies the heat-source model, track configuration, laser turnaround time, laser power, scan speed, hatch spacing, laser spot size, Double Conical Heat Source (DCHS) geometric and power-distribution parameters, and the heat transfer coefficients used in the model. This resource provides a consolidated reference for the simulation conditions and model parameters used throughout the study.

S4. Supporting graphs

Online Resource 24 (ESM_24.png).
Image showing the evolution of the predicted melt-pool dimensions with laser travel distance. The melt-pool length, width, and depth are plotted to demonstrate the development of a steady-state melt pool during single-track laser scanning. After the initial transient region, the melt-pool dimensions remain nearly constant with increasing laser travel distance, indicating that a steady-state condition has been reached. This analysis supports the selection of the track length used for evaluating the steady-state melt-pool dimensions in the simulations.

S5. PDFs

Online Resource 25 (ESM_25.pdf).
Image showing all type of mesh used. Showed detailed view for mesh used in single track cases AM-Bench 2022, Multi-track case AM-Bench 2022, Multi-track cases AM-Bench 2025.

Data Availability

The supplementary materials accompanying this article provide the numerical data, simulation videos, temperature histories, model parameters, and other supporting information used in the analyses presented in the manuscript as Online Resources 1–24.

The complete ANSYS Fluent simulation files used in this study, including the case and associated simulation files, are publicly available at:
https://drive.google.com/file/d/1emnQsmab7iHTuRSzj2tedGJ9wzvV-wEV/view?usp=sharing

Further data and simulation files related to this study are available from the corresponding author upon reasonable request.